

Assignment 2

1]STACK implementation using Array with PUSH, POP operations

Program -

```
class Stack
{
    constructor()
    {
        this.items = [];
    }

    add(element)
    {
        return this.items.push(element);
    }

    remove()
    {
        if(this.items.length > 0)
        {
            return this.items.pop();
        }
    }

    peek()
    {
        return this.items[this.items.length - 1];
    }

    isEmpty()
    {
        return this.items.length == 0;
    }

    size()
    {
        return this.items.length;
    }

    clear()
    {
        this.items = [];
    }
}

let stack = new Stack();
stack.add(1);
stack.add(2);
stack.add(4);
stack.add(8);
console.log(stack.items);
stack.remove();
console.log(stack.items);
console.log(stack.peek());
console.log(stack.isEmpty());
```

```
console.log(stack.size());  
stack.clear();  
console.log(stack.items);
```

Output -

```
[ 1, 2, 4, 8 ]  
[ 1, 2, 4 ]  
4  
false  
3  
[]
```